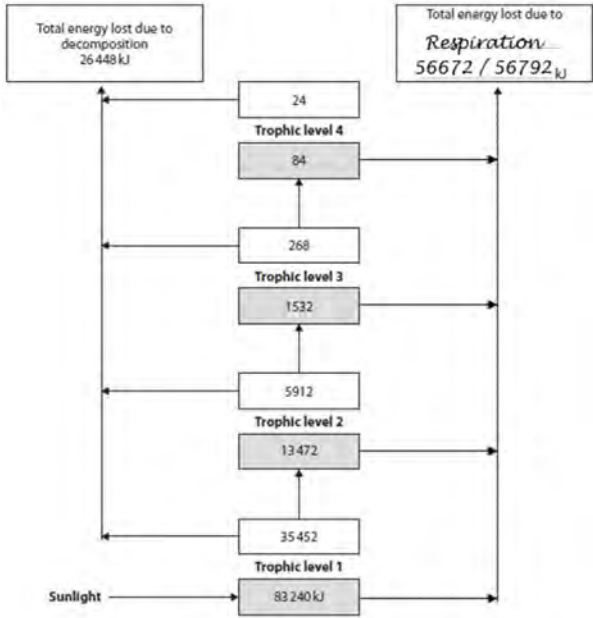


Question number	Answer	Additional guidance	Mark
8(a)(i)	An answer that includes the following points: • 35 452 and 83 240 (1) • 43 / 42.6 / 42.59 (%) (1)	Bald answer of {43 / 42.6 / 42.59} = 2 marks Bald answer containing {too many dps / incorrect rounding with acceptable number of dps} = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	An answer that includes the following points: • (total energy lost to) respiration / heat (energy) (1) • (total energy lost to) 56 672 / 56 792 (1) • four arrows each coming out of a different grey box pointing to the vertical RH arrow (1)	 The diagram illustrates energy flow through four trophic levels. At the base, 'Sunlight' provides energy to 'Trophic level 1' (83 240 kJ). This energy flows upwards through 'Trophic level 2' (35 452 kJ), 'Trophic level 3' (5 912 kJ), and 'Trophic level 4' (268 kJ). At each level, a grey box represents energy lost: 26 448 kJ at Trophic level 1, 13 472 kJ at Trophic level 2, 1 532 kJ at Trophic level 3, and 84 kJ at Trophic level 4. A vertical arrow on the right represents the total energy lost due to respiration, which is the sum of the losses from all four levels (26 448 + 13 472 + 1 532 + 84 = 56 792 kJ). The diagram also shows the total energy lost due to decomposition as 26 448 kJ.	(3)

Question number	Answer	Additional guidance	Mark
8(a)(iii)	An explanation that includes three of the following points: • because enzymes are needed to breakdown dead organisms (1) • so that the (products of) digested molecules are soluble (1) • so that the (digested) molecules can be {taken up / used} by microorganisms (1) • so that the dissolved molecules can soak into the ground (1) • important in {recycling / carbon cycle} (1)	ACCEPT named molecule e.g. cellulose ACCEPT named molecule e.g. glucose	(3)

Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes the following points: • (overall) {TLN / trophic level number} has increased (with time) (1) • because the (proportion of) meat in the diet has increased (1) • greatest increase in meat in diet was after_1983 (1) • people are eating less than 50% meat (throughout the time period) (1) • because the TLN is low (1)	ACCEPT proportion of plants has decreased ACCEPT people are eating more than 50% plants (in their diet) people eating more plants than animals / less meat than plants} ACCEPT below {3 / 2.5}	(3)

